Hero Adjusted Plus–Minus in *Marvel Rivals*

Intention-to-treat estimates from 477,483 competitive matches

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Raw win rates answer an easy question: which heroes are on the winning team most often? They do not answer the question we usually care about. Heroes are not randomly assigned to otherwise identical matches. Better players may select some heroes more often, and particular heroes may appear in stronger compositions, on favourable maps, or alongside specific team-ups. A 55% win rate therefore combines the effect of the hero with the circumstances in which that hero is chosen.

The object here is to separate the two as far as the data allow. Using 477,483 Season 9.5 competitive PC ranked matches, I estimate how the probability of winning changes when a team starts a particular hero rather than an average hero in the same role, holding fixed observed differences in map, team strength, role composition, and designated team-ups. I call this adjusted plus–minus, or APM.

A hero can have a high raw win rate because the hero is strong, because strong players and strong teams tend to select it, or both. The regression cannot eliminate every source of selection, but it can separate hero choice from several large, observable differences in match environment. The result is a different object from a leaderboard sorted by win rate.

Role composition

Role composition is the first identification problem. A hero coefficient should not receive credit for the fact that the hero happens to be played in a good role structure, or blame for being played in a bad one.

This is not a small issue in *Marvel Rivals*. The conventional 2–2–2 composition of two Tanks, two Damage heroes, and two Supports accounts for 75% of teams in the sample and wins 53.5% of its matches. A 1–3–2 composition wins 42.2%, and 2–1–3 wins 40.4%. These are large differences before anything has been said about which particular Tank, Damage, or Support heroes fill the slots.

I therefore include composition indicators directly. Each Tank–Damage–Support structure that appears at least 100 times receives its own coefficient, entered as the difference between the two teams, and the rare remainder is pooled into a single “other” category. This separates two sources of variation: composition coefficients compare role structures such as 2–2–2 and 1–3–2, while hero coefficients compare heroes within a role. Conditional on occupying a Tank slot, for example, The Thing is compared with the average Tank rather than being credited for the role structure around it.

Specification

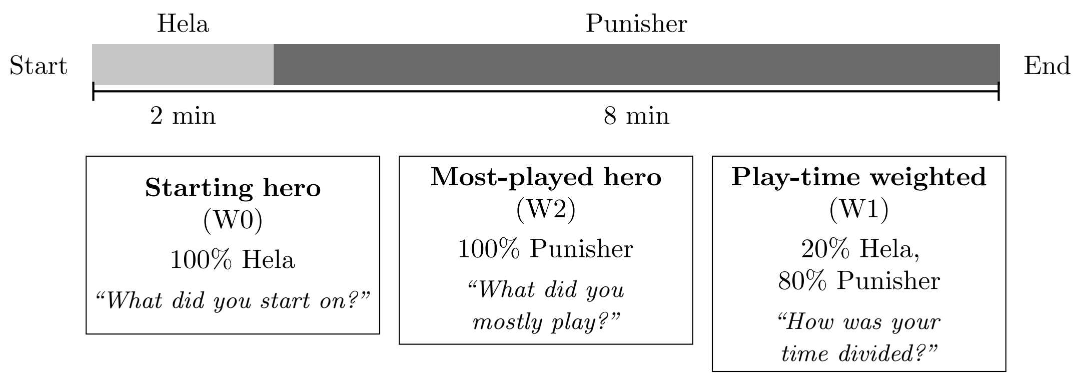
Each match is one observation and the outcome is whether side 0 wins. A hero enters as a signed contrast between the two teams: plus one if the hero starts on side 0, minus one if on side 1, and zero if on both or neither. The identifying variation for a hero coefficient is therefore the set of matches in which that hero appears on one side and not the other, compared after the other controls have been held fixed.

The remaining regressors absorb the obvious differences in match environment. Each of the 16 maps receives its own intercept, because side 0 wins between 49.42% and 52.52% of the time depending on the map. The pre-match rank-score difference between the teams controls for team strength. The composition contrasts described above and 100 designated team-up contrasts complete the specification. The model is an unpenalised logistic regression with HC1 heteroskedasticity-robust standard errors. Hero coefficients are normalised within role, for reasons given below, so every estimate is relative to the average hero of the same role.

Who gets credit when a player swaps?

Hero identity is not fixed within a match. Suppose a player starts Hela, plays her for two minutes, swaps to Punisher, and spends the remaining eight minutes there. There are three natural ways to encode that player. Starting-hero attribution (W0) counts the match as 100% Hela. Most-played-hero attribution (W2) counts it as 100% Punisher. Play-time weighting (W1) counts it as 20% Hela and 80% Punisher. Figure 1 lays the three side by side.

Figure 1: Three ways to credit one player who swaps heroes (illustrative match)



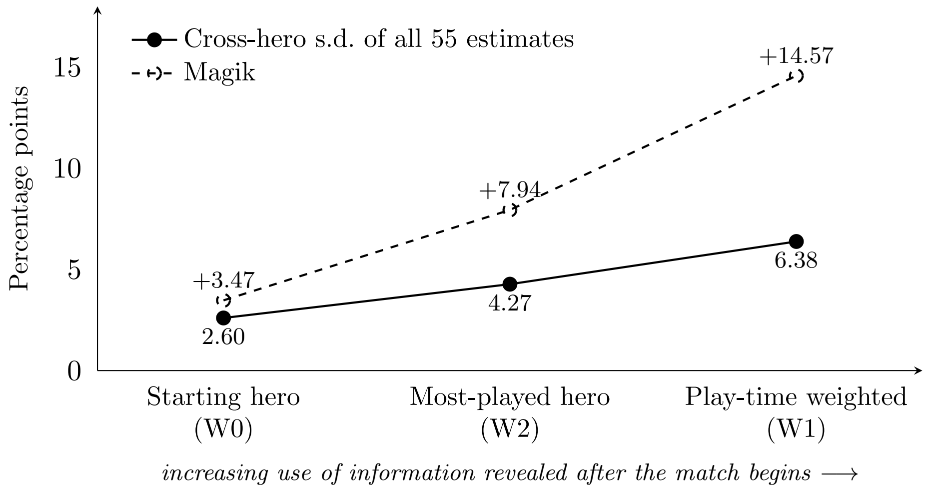
Notes. A stylised match. The three rules agree about any player who never swaps and disagree only about players who do, which is exactly where the outcome of the match can leak into the regressors. From left to right, each rule uses more information revealed after the match begins.

Play-time weighting uses more information, but some of that information is generated after the match begins. Players swap because a matchup is going badly, because the opposing team changes composition, because a hero stops working, or because the state of the game reveals information that was unavailable at the start. Realised hero minutes are therefore endogenous to the course of the match. Weighting hero exposure by those minutes gives the regression information about events it is supposed to explain, so a better in-sample fit under that rule says little about identification.

The data show exactly this pattern (Figure 2). The cross-hero standard deviation of estimated APM rises from 2.60 percentage points using starting heroes to 4.27 using the most-played hero and 6.38 using play-time weights. Magik moves from +3.47 to +7.94 to +14.57. Yet the rankings remain fairly similar, with rank correlations of 0.937 and 0.890 against the starting-hero ranking. What changes most is not who looks good, but how large the estimated effects become.

For the headline specification I therefore use the starting hero, accepting some measurement error in exchange for avoiding post-start information. The API records heroes chronologically by first appearance, so the starting choice is fixed before subsequent match developments can affect attribution. A player who abandons a hero after thirty seconds still counts as having started that hero. The resulting coefficient may be attenuated, but it has a clean intention-to-treat interpretation: what happens after a player begins the match on this hero? Appendix Table A4 reports every hero under all three rules.

Figure 2: The estimates expand as the attribution rule uses more post-start information



Notes. For each rule, filled circles show the cross-hero standard deviation of the 55 within-role estimates, all fitted on the same 477,483 matches. Open circles trace Magik, the hero whose estimate moves most between the starting-hero and play-time-weighted rules. Appendix Table A4 lists every hero under every rule.

Why hero effects are identified within role

Hero effects are identified within role, not across roles. The reason is mechanical. Changing the number of Tanks, Damage heroes, or Supports changes the team's composition, so a global normalisation asks the hero coefficients and the composition coefficients to explain nearly the same role-count variation.

There is no useful question in deciding whether an extra Tank belongs to the Tank coefficients or to the composition coefficient, so I impose the distinction instead. Tank effects sum to zero across Tanks, Damage effects across Damage heroes, and Support effects across Supports. Composition coefficients then measure role structure, and hero coefficients measure which heroes perform better or worse within a role.

Numerically, the correction changes very little. The resulting hero ranking has a 0.9999 rank correlation with the estimates under a single global normalisation, and the mean coefficient changes by only 0.010 percentage points. The reason for the within-role normalisation is therefore interpretation rather than fit: role structure belongs in the composition coefficients, while hero coefficients compare heroes occupying the same role.

Results

Table 1 reports every hero. Dispersion within role is substantial. Among Tanks, The Thing has the largest adjusted effect at 5.46 percentage points relative to the average Tank, followed by Magneto, Angela, Peni Parker, Hulk, and Thor. Among Damage heroes, Black Cat leads at 4.85 points, followed by Magik, Storm, Daredevil, and Mister Fantastic. Among Supports, Mantis stands apart at 6.76 points, followed by Rocket Raccoon, Ultron, Adam Warlock, and Loki. 10 of the 55 heroes cannot be distinguished from their role average after Benjamini–Hochberg false-discovery-rate correction.

These are marginal effects at a balanced match. The Thing's 5.46 is the change in predicted win probability from replacing an average Tank with The Thing, holding the controls fixed, evaluated where the two teams are otherwise even. It is an adjusted association rather than the causal effect of the pick, for reasons taken up below.

Alongside each APM the table reports the hero's starts and pick share within the crawl, its raw play-time-weighted win rate, and the change in the estimate when starting-hero attribution is replaced by play-time weighting. The raw win rate shows how far the adjustment moves each hero. The last column shows how far the attribution rule moves it. Those are different sources of distortion, and a reader can see which heroes are exposed to each.

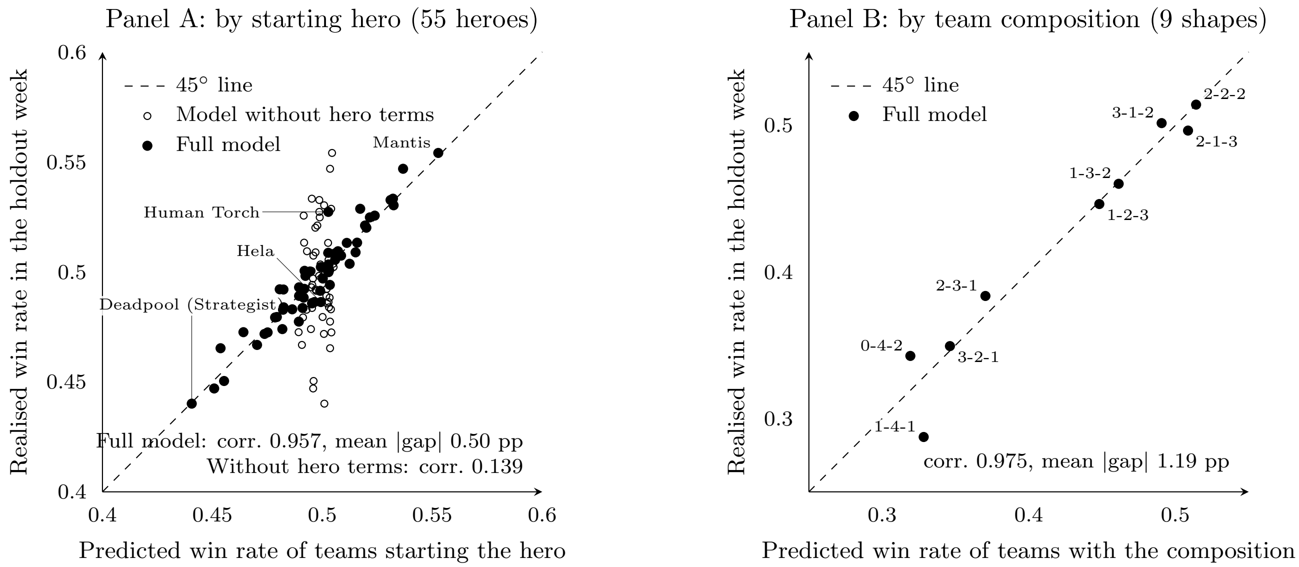
Does the structure survive out of sample?

With nearly half a million observations and 187 parameters, in-sample fit says relatively little. I instead test whether the estimated structure survives a chronological holdout.

I therefore estimate the model on the first 437,453 matches and evaluate it on the final 40,030, corresponding to the last 7 days of the sample. The split is chronological. A random split would make the test easier by mixing the same evolving meta across the training and test samples.

The model travels well (Figure 3). Across heroes, the predicted and realised holdout win rates of teams starting each hero have a correlation of 0.957 and differ by 0.50 percentage points on average. Removing the hero coefficients while keeping every other control reduces that correlation to 0.139. Across composition structures, predicted and realised win rates have a correlation of 0.975.

Figure 3: Does the model predict outcomes it has not seen? Holdout week, by hero and by composition



Notes. Fitted on the 437,453 matches before the final 7 days, evaluated on the 40,030 matches of that week. Each holdout match contributes two team-sides. Panel A groups them by each hero they started: the horizontal axis is the model's mean predicted win probability for those team-sides, the vertical axis the share that actually won. Filled circles use the full model. Open circles use the same model with the hero contrasts removed, so their horizontal spread is only what map, rank score, composition, and team-ups can explain about a hero's teams. Labels mark the two heroes farthest from the line and the two extremes. Panel B groups team-sides by composition shape (Tanks–Damage–Supports) with at least 100 holdout team-sides. Appendix Figure A1 gives the match-level calibration.

The holdout exercise is a persistence test, not an identification strategy. It shows that the hero coefficients retain information about later matches rather than merely fitting idiosyncrasies of the estimation sample.

The usual prediction diagnostics are consistent with the same result (Appendix Figure A1 and Table A2). Holdout log loss falls from 0.6928 for a constant forecast to 0.6674 for the full model, and the calibration slope is 1.067 with a standard error of 0.025. Adding the hero block to map, rank, and composition controls reduces holdout log loss from 0.6819 to 0.6698, so the hero terms add predictive information for subsequent matches.

Two further checks appear in Appendix Table A1. Randomly permuting hero assignments against outcomes removes 97.4% of the measured signal, so the estimates are not an artefact of the design matrix.

Correcting the role-composition problem raises the rank correlation between APM and raw hero win rate from 0.671 to 0.801, which is the expected direction: the adjusted estimates should disagree with raw win rates where the match environment differs across heroes, not everywhere.

Team-ups

The team-up coefficients (Appendix Table A6) estimate a different object. They ask whether a designated pair performs better or worse than the two heroes' individual effects would predict. Squirrel Girl with Iron Man carries a premium of 8.66 percentage points, Blade with Moon Knight 8.30, Rogue with Gambit 6.95, and Adam Warlock with Ultron 6.82. A premium is the increment beyond the two individual effects, so a team fielding two strong heroes with a positive team-up coefficient gains both hero terms plus the premium. Reading the premium as the total value of the pair would count the wrong object.

Designated team-ups are a small subset of the pairwise structure a match contains. A hero's value can depend on who it starts alongside and, just as plausibly, on who it starts against: some matchups are counters. I therefore estimate a second specification that adds one coefficient for every hero pair that starts together at least 2,500 times (a synergy, 1,115 pairs) and one for every pair that starts on opposite sides at least 2,500 times (a counter, 1,284 pairs), on the same starting lineups as the headline model. The blocks need constraints to be identified at all. Summing a hero's synergy contrasts over its five teammates gives five times its own contrast, and summing its counter contrasts over six opponents gives six times it, so without restrictions the interaction blocks contain the main effects exactly. Each hero's synergies are therefore constrained to sum to zero across partners and its counters to sum to zero across opponents, with a further sum-to-zero within each role pair so that composition stays in the composition block. The main effect then reads as a hero's value averaged over the partners and opponents it actually faces, and each interaction is a deviation from that average.

The main effects change definition rather than evidence. Under the headline specification a hero's coefficient is net of its designated team-up premiums; here it averages over every partner the hero actually starts with, team-ups included. The rank correlation with the headline estimates is 0.869 and the mean absolute change 1.07 percentage points, and the changes track team-up exposure (correlation 0.45 with the sum of a hero's positive team-up premiums): Star-lord (+2.6), Cyclops (+2.6), and Peni Parker (+2.2) gain, while Daredevil (-3.0), Magik (-2.6), and The Thing (-2.3) lose. Neither version is wrong. They answer different questions about the same hero.

After false-discovery-rate correction across all 2,399 interactions, 52 counters and 74 synergies are distinguishable from zero. The largest synergies are Moon Knight with Blade (+8.1), Ultron with Adam Warlock (+7.7), and Squirrel Girl with Iron Man (+7.2), all of them designated team-ups. Of the 100 designated team-ups, 32 are significant, against 42 of the 1,015 other pairs, which is the pattern one would expect if the game's designed synergies are the main synergies. The largest counters are Black Widow over Ultron (+4.8), Ultron over White Fox (+4.7), and Deadpool (Duelist) over Deadpool (Strategist) (+3.9). Two folk beliefs can be checked directly. Black Panther against a starting Thing is -2.2 points (standard error 0.8, $q = 0.05$, 7,099 matches), so most of the raw -5.6-point gap in Black Panther's win rate when a Thing is on the other side is The Thing's own strength rather than a specific counter. Jeff starting alongside Devil Dinosaur is +1.9 points (standard error 0.8, $q = 0.18$, 7,082 team instances). On the chronological holdout the interaction blocks do not improve the holdout log loss, which moves from 0.6674 under the headline specification to 0.6677 with the synergy block and 0.6682 with both blocks. Individual interactions can be precisely estimated while the blocks as a whole, with 2,374 free parameters, add more noise than signal to prediction. The headline model therefore keeps the parsimonious team-up block, and the interactions are reported as a description of the sample rather than as part of the estimate (Appendix Table A3). Appendix Table A5 lists the strongest interactions of each kind.

What the coefficients do and do not mean

The coefficient is not hero strength in any structural or causal sense. Three qualifications matter.

First, magnitudes depend on the attribution rule. Rankings are fairly stable across the starting-hero, most-played, and play-time specifications, but the dispersion of the coefficients is not, at 2.60, 4.27, and 6.38 percentage points respectively. I therefore place substantially more confidence in statements such as "Magik looks strong relative to other Damage heroes" than in statements such as "Magik is worth exactly +3.47 percentage points."

Second, hero choice remains endogenous. Rank score controls for broad differences in player strength, but not for hero-specific skill. A player with five hundred hours on Magik is not observationally equivalent to a player selecting Magik for the first time. Without player-by-hero proficiency, that source of selection remains in the coefficient.

Third, the reported standard errors are likely optimistic. HC1 treats matches as independent observations, while players recur across matches and twelve players appear in each match, so the dependence structure is cross-classified. The point estimates do not depend on this issue, but inference does. A player-cluster bootstrap would address it and has not yet been run, so the confidence intervals should be read accordingly.

The data also thin out at the edges. 18 composition shapes receive their own indicators, while 9 extremely rare shapes, together only 0.024% of team instances, share the "other" category and one coefficient. Profile privacy rises sharply with rank score, reaching roughly 63% above 5,000, and the crawl expands only through public profiles, so the very top of the ranked distribution is under-sampled. At this sample size the binding constraint is coverage at the top rather than sampling noise.

Conclusion

Raw win rates combine hero choice with the environment in which that choice is made. Adjusting for map, team strength, role composition, and designated team-ups changes those comparisons materially and leaves persistent differences across heroes within each role.

I use starting-hero attribution because it avoids conditioning hero identity on information generated after the match begins. The estimates should therefore be read as intention-to-treat adjusted associations: conditional on the observed match environment, how does winning change when a team starts this hero rather than an average hero in the same role?

I place more weight on the ranking than on the exact coefficient magnitudes. Rankings are comparatively stable across attribution rules, while the scale of the estimates is not.

Table 1: Hero adjusted plus–minus, relative to an average hero of the same role

	APM	S.E.	q	Starts	Pick (%)	Raw WR (%)	Play-time − Starting		APM	S.E.	q	Starts	Pick (%)	Raw WR (%)	Play-time − Starting
Panel A: Tank (15 heroes)								*Panel B: Damage (27 heroes)*							
The Thing	5.46***	(0.204)	0.000	122,883	2.12	52.4	+1.75	Black Cat	4.85***	(0.260)	0.000	66,494	1.15	55.2	+6.77
Magneto	1.96***	(0.139)	0.000	400,170	6.90	50.9	+1.60	Magik	3.47***	(0.219)	0.000	153,326	2.65	56.8	+11.11
Angela	1.87***	(0.261)	0.000	45,696	0.79	51.8	+0.81	Storm	3.09***	(0.305)	0.000	45,132	0.78	55.3	+7.18
Peni Parker	1.51***	(0.328)	0.000	43,799	0.76	54.6	+5.90	Daredevil	2.92***	(0.217)	0.000	75,418	1.30	53.6	+5.55
Hulk	1.33***	(0.219)	0.000	81,337	1.40	52.3	+2.21	Mister Fantastic	2.52***	(0.351)	0.000	40,940	0.71	50.4	+1.43
Thor	1.29***	(0.219)	0.000	80,452	1.39	50.4	-0.64	Iron Man	1.34***	(0.279)	0.000	50,391	0.87	51.5	+1.76
The Hood	0.45**	(0.172)	0.012	404,062	6.97	52.8	+6.08	Iron Fist	1.32***	(0.224)	0.000	85,109	1.47	52.1	+3.19
Devil Dinosaur	-0.03	(0.231)	0.907	69,629	1.20	48.6	-2.21	Hela	1.14***	(0.199)	0.000	110,771	1.91	49.8	+0.08
Groot	-0.67**	(0.290)	0.027	58,919	1.02	49.9	-3.13	Winter Soldier	1.00***	(0.141)	0.000	228,817	3.95	49.3	-0.83
Venom	-0.93***	(0.219)	0.000	103,844	1.79	50.2	-0.41	Wolverine	0.82**	(0.317)	0.012	42,210	0.73	49.3	-0.61
Doctor Strange	-0.98***	(0.202)	0.000	135,405	2.34	48.4	-5.85	Spider-man	0.58***	(0.175)	0.001	142,393	2.46	50.9	+2.32
Captain America	-1.22***	(0.352)	0.001	33,294	0.57	50.4	-0.57	Moon Knight	0.10	(0.297)	0.792	54,845	0.95	49.0	-3.47
Emma Frost	-1.77***	(0.192)	0.000	167,701	2.89	46.9	-4.91	Black Widow	0.02	(0.200)	0.907	84,335	1.46	48.7	-1.26
Rogue	-3.41***	(0.344)	0.000	52,686	0.91	51.0	+1.64	Blade	-0.20	(0.317)	0.578	47,659	0.82	49.8	+1.28
Deadpool (Vanguard)	-4.86***	(0.252)	0.000	40,623	0.70	47.9	-2.28	Black Panther	-0.21	(0.239)	0.423	65,701	1.13	50.4	+3.08
Panel C: Support (13 heroes)								Human Torch	-0.28	(0.351)	0.468	55,560	0.96	54.1	+4.84
Mantis	6.76***	(0.181)	0.000	148,036	2.55	56.9	+6.74	Elsa Bloodstone	-0.39	(0.229)	0.108	89,462	1.54	50.1	+0.26
Rocket Raccoon	4.14***	(0.150)	0.000	198,679	3.43	52.5	+1.03	Psylocke	-0.58	(0.342)	0.106	46,440	0.80	50.3	+2.12
Ultron	3.59***	(0.297)	0.000	39,543	0.68	53.3	+3.17	Deadpool (Duelist)	-0.67***	(0.220)	0.003	57,041	0.98	51.5	+2.52
Adam Warlock	2.08***	(0.315)	0.000	40,028	0.69	51.5	+1.84	Namor	-0.82***	(0.243)	0.001	75,351	1.30	45.2	-6.75
Loki	1.18***	(0.268)	0.000	48,317	0.83	51.3	+0.66	Hawkeye	-1.22***	(0.341)	0.001	40,267	0.69	48.8	-1.03
Cloak & Dagger	-0.04	(0.170)	0.852	320,390	5.53	49.9	+0.42	The Punisher	-1.81***	(0.262)	0.000	75,297	1.30	46.2	-8.64
Gambit	-0.30	(0.210)	0.178	195,439	3.37	49.9	-0.34	Star-lord	-2.41***	(0.300)	0.000	51,580	0.89	49.7	-1.64
Invisible Woman	-1.38***	(0.151)	0.000	228,763	3.95	48.0	-4.01	Scarlet Witch	-2.49***	(0.414)	0.000	46,118	0.80	46.1	-3.43
Jeff The Land Shark	-1.75***	(0.233)	0.000	71,748	1.24	46.8	-3.27	Squirrel Girl	-3.78***	(0.350)	0.000	32,402	0.56	42.9	-12.83
Jubilee	-1.91***	(0.172)	0.000	290,670	5.02	50.5	+2.23	Cyclops	-4.04***	(0.195)	0.000	108,037	1.86	45.2	-3.61
Luna Snow	-3.01***	(0.167)	0.000	197,630	3.41	46.6	-4.24	Phoenix	-4.28***	(0.345)	0.000	45,826	0.79	44.0	-9.38
White Fox	-3.48***	(0.208)	0.000	99,670	1.72	46.1	-3.21								
Deadpool (Strategist)	-5.89***	(0.210)	0.000	59,610	1.03	45.5	-1.02								
Matches	477,483							Log-likelihood	-320,067.7						
Player–matches	5,729,796							Heroes	55						

Notes. All 55 heroes are listed. *APM* is in percentage points of win probability at a balanced match ($\partial p/\partial x = \beta/4$): the change from fielding this hero at match start in place of an average hero of the same role. Standard errors in parentheses are HC1, and the 95% confidence interval is APM $\pm$ 1.96 S.E. Stars and q denote Benjamini–Hochberg false-discovery-rate control across the 55-hero family: * $q < 0.10$, ** $q < 0.05$, *** $q < 0.01$. Of the 55 heroes, 10 are not distinguishable from their role average. *Starts* is the number of player-slots that began the match on this hero and *Pick %* its share of all starting slots. Both reflect the crawl's per-hero leaderboard seeding, not population pick rates. *Raw WR* is the play-time-weighted raw win rate, an assumption-light benchmark that controls for nothing. *Play-time − Starting* (W1−W0) is how far play-time-weighted attribution inflates the estimate relative to starting-hero attribution, in points. Sample: 482,997 matches crawled, less 2,139 containing draws (`is_win` = 2), 19 with incomplete play time, and 3,356 below a 240-second forfeit floor. Four team-up contrasts were structurally empty and dropped. Appendix Table A4 repeats every hero under all three attribution rules.

Appendix

Table A1: Specification checks

Check	Result	Interpretation
Placebo (permuted heroes)	0.023 vs 0.881	97.4% of signal destroyed
Sum-to-zero normalisation	within-role	holds at machine precision
Rank agreement, raw win rates	$\rho = 0.801$	up from 0.671 before correction
Within-role reparameterisation	$\rho = 0.9999$	mean change 0.010 pp
Camp indexing	49.87% camp 0	absolute map side, no crawl selection
Temporal holdout (7 days)	log loss 0.6674	vs 0.6928 constant, slope 1.067

The placebo shuffles hero assignment against outcomes and refits. A placebo that failed would invalidate the specification. Camp-0 win rate varies 49.42%–52.52% across the 16 maps, which is why each map carries its own intercept. The holdout row summarises Figure A1.

Table A2: Out-of-sample fit as blocks of the model are added

Model	Parameters	Log-lik. (train)	Log loss (train)	Log loss (holdout)
Map intercepts only	16	−303,099	0.6929	0.6927
+ rank-score differential	17	−301,826	0.6900	0.6874
+ team-composition shapes	35	−299,416	0.6845	0.6819
+ hero contrasts	87	−294,223	0.6726	0.6698
+ team-up contrasts	187	−293,362	0.6706	0.6674

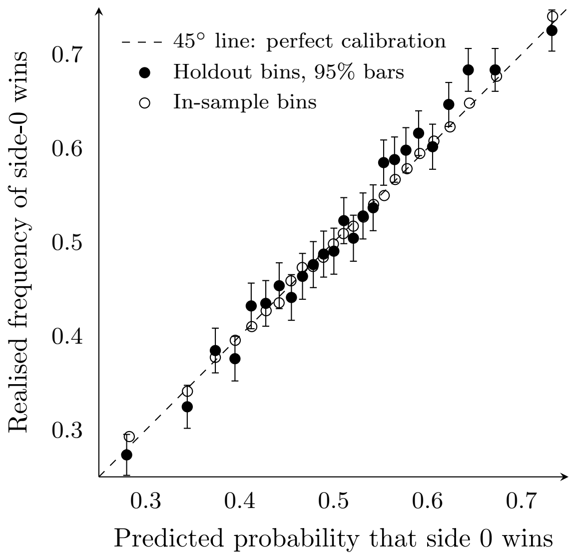
Each row adds one block to the model in the row above. The last row is the headline specification. Fitted on the 437,453 matches before the final 7 days, evaluated on the 40,030 matches of that week. A constant prediction at the training base rate gives a holdout log loss of 0.6928. The in-sample column shows how little of the improvement is overfitting.

Table A3: Out-of-sample fit with pairwise interactions

Model	Free parameters	Log loss (holdout)
Spec A+ (heroes, maps, skill, shapes, team-ups)	187	0.6674
+ all pairwise synergies (team-ups subsumed)	1144	0.6677
+ all pairwise counters	2374	0.6682

Same chronological split as Table A2. The synergy block replaces the designated team-up block, which it contains; pairs below 2,500 co-occurrences carry no coefficient. Free parameters count the coefficients remaining after the identification constraints.

Figure A1: Out-of-sample calibration of the headline model



Holdout, last 7 days, $N = 40,030$

Log loss	0.6674
constant prediction	0.6928
Brier score	0.2374
AUC	0.627
Calibration slope (s.e.)	1.067 (0.025)
Calibration intercept (s.e.)	0.018 (0.010)

Notes. The model is fitted on the 437,453 matches played before the final 7 days of the sample and evaluated on the 40,030 matches played during them, since a random split would leak meta drift. Matches are sorted by predicted probability into 25 equal-count bins of about 1,600 matches. Each filled circle is one bin's mean prediction against its realised win frequency, with a 95% binomial interval. Open circles repeat the construction in-sample on all 477,483 matches. The calibration slope and intercept are from a logit of the realised outcome on the log-odds of the prediction (ideal: 1 and 0). A constant prediction at the training base rate gives the log loss shown for comparison. Predicted probabilities fall between 5% and 93%, and 85% of holdout matches are predicted between 35% and 65%: the model separates unbalanced matches well and, as it should, treats balanced ones as coin flips.

Table A4: What happens to estimated hero strength when swaps affect attribution?

	more post-start information ⟶					*more post-start information* ⟶			
	Starting hero (W0)	Most-played hero (W2)	Play-time weighted (W1)	Difference (W1−W0)		Starting hero (W0)	Most-played hero (W2)	Play-time weighted (W1)	Difference (W1−W0)
Panel A: Tank (15 heroes)					*Panel B: Damage (27 heroes)*				
The Thing	5.46	4.94	7.21	+1.75	Black Cat	4.85	7.74	11.62	+6.77
Magneto	1.96	0.55	3.57	+1.60	Magik	3.47	7.94	14.57	+11.11
Angela	1.87	2.00	2.68	+0.81	Storm	3.09	6.57	10.27	+7.18
Peni Parker	1.51	10.18	7.41	+5.90	Daredevil	2.92	5.70	8.47	+5.55
Hulk	1.33	2.03	3.55	+2.21	Mister Fantastic	2.52	3.02	3.95	+1.43
Thor	1.29	0.73	0.65	-0.64	Iron Man	1.34	2.21	3.10	+1.76
The Hood	0.45	0.35	6.52	+6.08	Iron Fist	1.32	2.84	4.52	+3.19
Devil Dinosaur	-0.03	-0.56	-2.24	-2.21	Hela	1.14	1.04	1.22	+0.08
Groot	-0.67	-1.52	-3.79	-3.13	Winter Soldier	1.00	-0.07	0.17	-0.83
Venom	-0.93	-1.25	-1.35	-0.41	Wolverine	0.82	0.62	0.21	-0.61
Doctor Strange	-0.98	-2.55	-6.82	-5.85	Spider-man	0.58	1.48	2.90	+2.32
Captain America	-1.22	-0.51	-1.78	-0.57	Moon Knight	0.10	-2.01	-3.37	-3.47
Emma Frost	-1.77	-4.62	-6.68	-4.91	Black Widow	0.02	-0.96	-1.24	-1.26
Rogue	-3.41	-3.98	-1.78	+1.64	Blade	-0.20	0.28	1.09	+1.28
Deadpool (Vanguard)	-4.86	-5.79	-7.15	-2.28	Black Panther	-0.21	1.20	2.87	+3.08
Panel C: Support (13 heroes)					Human Torch	-0.28	1.74	4.56	+4.84
Mantis	6.76	9.88	13.50	+6.74	Elsa Bloodstone	-0.39	-0.46	-0.12	+0.26
Rocket Raccoon	4.14	4.50	5.18	+1.03	Psylocke	-0.58	0.08	1.54	+2.12
Ultron	3.59	6.92	6.76	+3.17	Deadpool (Duelist)	-0.67	0.48	1.85	+2.52
Adam Warlock	2.08	4.11	3.93	+1.84	Namor	-0.82	-3.87	-7.57	-6.75
Loki	1.18	2.14	1.84	+0.66	Hawkeye	-1.22	-1.31	-2.25	-1.03
Cloak & Dagger	-0.04	-1.84	0.38	+0.42	The Punisher	-1.81	-5.38	-10.45	-8.64
Gambit	-0.30	-2.07	-0.64	-0.34	Star-lord	-2.41	-2.64	-4.05	-1.64
Invisible Woman	-1.38	-3.32	-5.39	-4.01	Scarlet Witch	-2.49	-4.65	-5.91	-3.43
Jeff The Land Shark	-1.75	-3.04	-5.01	-3.27	Squirrel Girl	-3.78	-8.62	-16.61	-12.83
Jubilee	-1.91	-1.94	0.32	+2.23	Cyclops	-4.04	-5.45	-7.65	-3.61
Luna Snow	-3.01	-5.00	-7.25	-4.24	Phoenix	-4.28	-7.54	-13.67	-9.38
White Fox	-3.48	-4.73	-6.69	-3.21					
Deadpool (Strategist)	-5.89	-5.62	-6.91	-1.02					
Log-likelihood	−320,122	−305,869	−297,648		Cross-hero s.d.	2.60	4.27	6.38	
Rank corr. with starting hero	—	0.937	0.890						

Notes. Starting-hero attribution (W0) assigns the match to the hero selected at the start. Most-played-hero attribution (W2) assigns it to whichever hero the player used longest. Play-time weighting (W1) splits attribution according to realised play time. All three specifications use the same 477,483 matches, and estimates are within-role, in percentage points. Moving from W0 toward W1 gives the regression progressively more information about what happened after the match began. The coefficients become much larger, with cross-hero dispersion rising by a factor of 2.5, while the broad hero ranking stays comparatively stable. The difference (W1−W0) is reported rather than a ratio because a ratio diverges for heroes whose starting-hero estimate is near zero. The better in-sample likelihood under W1, −297,648 against −320,122, is not evidence that W1 identifies hero strength better; part of the realised match has been placed inside the regressors.

Table A5: Strongest pairwise interactions on starting lineups

Counters: A over B				*Synergies: A with B*			
Matchup	Effect	S.E.	Matches	Pair	Effect	S.E.	Teams
Black Widow over Ultron	+4.82***	(1.01)	3,256	Moon Knight × Blade	+8.09***	(0.85)	4,378
Ultron over White Fox	+4.68***	(0.99)	4,228	Ultron × Adam Warlock	+7.66***	(0.75)	6,510
Deadpool (Duelist) over Deadpool (Strategist)	+3.87**	(1.07)	3,133	Squirrel Girl × Iron Man	+7.17***	(0.94)	3,394
Venom over Squirrel Girl	+3.78***	(0.97)	3,421	Iron Fist × White Fox	+6.76***	(0.63)	15,235
Thor over Black Panther	+3.67***	(0.82)	5,728	Gambit × Rogue	+6.51***	(0.64)	26,196
Storm over Iron Fist	+3.57***	(0.90)	3,986	Loki × Mantis	+4.73***	(0.84)	7,635
Phoenix over Elsa Bloodstone	+3.34***	(0.91)	3,859	Mantis × Adam Warlock	+4.00***	(0.84)	4,710
Spider-man over Iron Man	+3.31***	(0.74)	6,082	The Punisher × Rocket Raccoon	+3.98***	(0.58)	24,911
Black Widow over Iron Man	+3.18**	(0.88)	4,090	Luna Snow × White Fox	+3.80***	(0.66)	18,249
Peni Parker over Thor	+3.08**	(0.98)	3,726	Storm × Jeff The Land Shark	+3.79***	(0.73)	8,014
White Fox over Wolverine	+2.97**	(0.93)	3,959	Ultron × Devil Dinosaur	-3.62**	(1.10)	2,672
Magik over Devil Dinosaur	+2.93***	(0.64)	11,037	Rocket Raccoon × Invisible Woman	-3.62***	(0.57)	28,719
Gambit over Angela	+2.91***	(0.79)	9,187	Rocket Raccoon × Jeff The Land Shark	-3.59***	(0.77)	7,951
Hela over Ultron	+2.84**	(0.90)	4,310	Doctor Strange × Invisible Woman	+3.53***	(0.52)	45,963
The Thing over Rogue	+2.71**	(0.83)	6,170	Hela × Iron Man	-3.49**	(0.99)	2,852
Daredevil over Black Cat	+2.70**	(0.85)	4,640	Rocket Raccoon × Gambit	-3.43***	(0.60)	22,783
Namor over Spider-man	+2.68***	(0.63)	9,237	Mantis × Peni Parker	-3.39***	(0.81)	5,603
Black Panther over Mantis	+2.58**	(0.75)	10,240	Human Torch × Jubilee	+3.34***	(0.60)	21,864
The Thing over Deadpool (Duelist)	+2.58**	(0.78)	6,657	Gambit × Deadpool (Duelist)	+3.32***	(0.65)	15,503
Cloak & Dagger over Cyclops	+2.55***	(0.57)	36,351	Adam Warlock × Elsa Bloodstone	-3.31**	(1.00)	3,209
Rocket Raccoon over Squirrel Girl	+2.53**	(0.81)	6,380	Loki × Devil Dinosaur	+3.30**	(1.03)	3,444
Gambit over Wolverine	+2.51**	(0.74)	9,067	Adam Warlock × Cyclops	-3.22**	(0.91)	3,993
Devil Dinosaur over Magneto	+2.41***	(0.57)	26,012	Doctor Strange × Angela	-3.12**	(0.99)	3,344
Winter Soldier over Ultron	+2.41**	(0.69)	9,083	Hela × Ultron	-3.01**	(0.95)	3,497
Deadpool (Duelist) over Invisible Woman	+2.39**	(0.71)	13,683	Namor × Gambit	-2.99***	(0.64)	14,862
Iron Man over The Thing	+2.38**	(0.77)	6,517	Rocket Raccoon × Cloak & Dagger	-2.98***	(0.53)	40,371
Human Torch over Jubilee	+2.28**	(0.68)	16,056	Adam Warlock × Jubilee	+2.98***	(0.76)	6,613
The Hood over Storm	+2.28***	(0.60)	18,350	Cloak & Dagger × Peni Parker	-2.97***	(0.66)	13,400
Angela over Magneto	+2.25**	(0.62)	18,811	Cloak & Dagger × Invisible Woman	-2.92***	(0.53)	43,646
The Thing over Magik	+2.21***	(0.53)	17,714	Rocket Raccoon × Jubilee	-2.92***	(0.53)	34,943

Notes. The 30 largest counters and the 30 largest synergies among those significant at $q < 0.05$ after Benjamini–Hochberg correction across all 2,399 interactions (52 counters and 74 synergies are significant in total). A counter of $+x$ means that when A starts against B, A's side wins x percentage points more often than the two heroes' individual effects predict, at a balanced match. A synergy of $+x$ means a team starting both A and B wins x points more than their individual effects predict. Interactions are deviations: each hero's counters sum to zero across its opponents and its synergies across its partners, so a hero's main effect already includes its average matchup. HC1 standard errors in parentheses. Full tables: `results/apm_pairwise_counter_W0_specE.csv` and `results/apm_pairwise_synergy_W0_specE.csv`.

Table A6: Team-up synergies, all 100 pairs

Team-up	Premium	S.E.	Team-up	Premium	S.E.	Team-up	Premium	S.E.
Squirrel Girl × Iron Man	8.66	(0.965)	White Fox × Black Cat	1.77	(0.587)	Loki × Hela	0.45	(0.708)
Blade × Moon Knight	8.30	(0.851)	Black Panther × Storm	1.77	(0.996)	Hela × Namor	0.21	(0.552)
Rogue × Gambit	6.95	(0.472)	Hela × Venom	1.74	(0.552)	Gambit × Jubilee	0.21	(0.322)
Adam Warlock × Ultron	6.82	(0.768)	Thor × Hela	1.71	(0.552)	Mister Fantastic × The Thing	0.17	(0.773)
Iron Fist × White Fox	6.61	(0.498)	Venom × Phoenix	1.67	(0.596)	Luna Snow × Adam Warlock	0.16	(0.763)
Mantis × Star-lord	4.27	(0.503)	Mister Fantastic × Rocket Raccoon	1.62	(0.591)	Devil Dinosaur × The Punisher	0.10	(0.700)
The Punisher × Rocket Raccoon	4.13	(0.422)	Blade × Captain America	1.62	(0.981)	Namor × Luna Snow	0.09	(0.483)
Storm × Jeff The Land Shark	3.86	(0.667)	The Punisher × Daredevil	1.58	(0.850)	Spider-man × Venom	0.06	(0.424)
Human Torch × Jubilee	3.71	(0.469)	Hawkeye × Cloak & Dagger	1.58	(0.542)	Groot × Jeff The Land Shark	-0.02	(0.819)
Loki × Mantis	3.53	(0.667)	Ultron × Iron Man	1.54	(0.864)	Psylocke × Cloak & Dagger	-0.03	(0.514)
Doctor Strange × Hulk	3.47	(0.603)	Emma Frost × Mantis	1.50	(0.363)	Venom × Blade	-0.04	(0.750)
Magik × The Hood	3.36	(0.295)	Daredevil × Black Widow	1.48	(0.728)	Captain America × Thor	-0.06	(1.079)
Adam Warlock × Storm	3.28	(1.149)	Wolverine × Gambit	1.47	(0.581)	Moon Knight × Elsa Bloodstone	-0.06	(0.940)
Luna Snow × White Fox	3.19	(0.447)	Magik × Doctor Strange	1.45	(0.389)	Winter Soldier × The Punisher	-0.07	(0.538)
Peni Parker × Rocket Raccoon	3.05	(0.543)	Jeff The Land Shark × Venom	1.41	(0.600)	Magneto × Emma Frost	-0.12	(0.313)
Cyclops × Phoenix	2.79	(0.641)	Angela × Star-lord	1.39	(1.006)	Black Cat × Black Panther	-0.25	(1.018)
Mantis × Adam Warlock	2.76	(0.831)	Angela × Loki	1.35	(0.957)	Jubilee × Blade	-0.41	(0.519)
Cloak & Dagger × The Hood	2.60	(0.233)	Invisible Woman × Human Torch	1.32	(0.507)	Invisible Woman × Mister Fantastic	-0.44	(0.614)
Doctor Strange × Invisible Woman	2.58	(0.326)	Gambit × Magneto	1.31	(0.277)	The Thing × Human Torch	-0.45	(0.548)
Scarlet Witch × Jubilee	2.47	(0.510)	Star-lord × Groot	1.24	(0.784)	Psylocke × Emma Frost	-0.49	(0.615)
Jubilee × The Hood	2.46	(0.236)	Elsa Bloodstone × Devil Dinosaur	1.21	(0.583)	Magneto × Scarlet Witch	-0.55	(0.511)
Black Widow × Phoenix	2.44	(0.859)	The Thing × Invisible Woman	1.20	(0.358)	Rocket Raccoon × Groot	-0.55	(0.518)
Hulk × Captain America	2.39	(0.998)	Rogue × Magneto	1.07	(0.501)	Thor × Angela	-0.81	(1.007)
Hawkeye × Psylocke	2.35	(0.907)	Cloak & Dagger × Luna Snow	1.07	(0.318)	Iron Man × Hulk	-0.86	(0.796)
Squirrel Girl × Spider-man	2.17	(0.959)	Captain America × Winter Soldier	0.98	(0.672)	Cyclops × Gambit	-1.00	(0.411)
Ultron × Peni Parker	2.11	(1.072)	Namor × Hulk	0.86	(0.585)	Storm × Thor	-1.17	(0.733)
Spider-man × Peni Parker	2.02	(0.714)	Devil Dinosaur × Jeff The Land Shark	0.83	(0.679)	Hulk × Wolverine	-1.39	(0.805)
Winter Soldier × Elsa Bloodstone	2.00	(0.442)	Phoenix × Rogue	0.79	(1.066)	Black Widow × Hawkeye	-1.54	(1.126)
Phoenix × Hela	1.98	(0.808)	Iron Fist × The Thing	0.73	(0.490)	Daredevil × Iron Fist	-1.80	(0.741)
Moon Knight × Cloak & Dagger	1.97	(0.465)	Emma Frost × Luna Snow	0.70	(0.332)	Human Torch × Storm	-1.80	(1.057)
Star-lord × Adam Warlock	1.95	(1.190)	Wolverine × Cyclops	0.68	(0.881)	Black Panther × Magik	-2.09	(0.666)
Iron Man × Thor	1.87	(0.688)	White Fox × Psylocke	0.61	(0.778)	Black Cat × Spider-man	-2.24	(0.627)
Scarlet Witch × Doctor Strange	1.85	(0.628)	Peni Parker × Black Panther	0.52	(1.041)			
Groot × Mantis	1.84	(0.535)	Rocket Raccoon × Squirrel Girl	0.50	(0.726)			

Notes. All 100 team-ups are listed in descending order of premium, reading down each column in turn. Each is labelled by its member heroes, anchor first. The premium is what a pair earns *beyond* what its two heroes contribute individually, in percentage points, under starting-hero attribution, with HC1 standard errors in parentheses. Four further team-ups are absent because they are defined against hero id 1057, base “Deadpool”, whose plays are all recorded under his three role variants, leaving those columns structurally empty.